

**DEVELOPMENT OF THERMOELECTRIC GENERATOR
DEMONSTRATOR UNIT**

TECHNICAL DISPLAY

Developed by:

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For:

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Fall 2026

Proposal

The outlined project will detail and document the creation of small Thermoelectric Generator Demonstration unit (TGDU) for the purposes of showing working principles. Principles demonstrated will include; Conversion of heat to electricity, circuit design, and system calculations. Fundamentally the conversion of heat to electricity is what powers Radioisotope generators and enables space missions beyond Earth's orbit. The demonstration will make use of common components such as Peltier modules, porcelain cup and simple electronics whose purpose will be to power a small LED to demonstrate the heat-to-electricity conversion. This project will fulfill the requirements set forth by the term project and demonstrate knowledge of concepts learned in class.

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1. Requirements

Requirements for the term technical project shall consist of the following to be carried out in full by 21st of April, 2026.

- The “system” must maintain a temperature difference to demonstrate operating principles.
- Electrical output must generate a load to power a small indicator separate of outside sources; such as, LED light, voltmeter or a small electric motor to show operating principles.
 - Creation of a “boost” circuit to power these devices shall be provided and documented in the appropriate section of the final report.
- System calculations shall be documented, and physical measurements of all relevance should be included in a final report, such measurements can include;
 - Temperature of outside container as a function of time.
 - Current produced by Peltier modules as a function of time.
- Final documentation and technical drawings shall be retained and included in the final report ensuring all results and drawings are verifiable to a complete system.

2. Component selection

- Peltier modules
- Aluminum heat sinks
- Thermal paste
- Kapton tape
- 2N3904 or PN2222 NPN transistor
- 1k ohm resistor
- Ferrite toroid core
- Enameled magnet wire
- Electrolytic capacitors
- LED
- Schottky diode
- Raspberry Pi RP2350B
- K-Type thermocouples
- MAX31855 or MAX6675 interface modules
- INA219 current/voltage sensor
- Breadboard

3. Experimental setup

